Q1.

1.1

(a)

We know Essential matrix is:

$$\mathcal{E} = [t]_\times R$$

NOTE: $[x]_\times = \begin{bmatrix} 0 & -z & y \\ z & 0 & -x \\ -y & x & 0 \end{bmatrix}$

As, $R = I$, we can ignore it.

(i) $\mathcal{E}_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$

Epipoles:

$$\mathcal{E} e_1 = 0 \qquad\qquad e_2^T \mathcal{E} = 0$$

$$\mathcal{E}^T e_2 = 0$$

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \underline{0} \qquad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \underline{0}$$

$$\left[\begin{array}{ccc:c} 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 \end{array}\right] R_1 \leftrightarrow R_3 \qquad \left[\begin{array}{ccc:c} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 \end{array}\right] R_1 \leftrightarrow R_3$$

$$\left[\begin{array}{ccc:c} 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \qquad \left[\begin{array}{ccc:c} 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \qquad\qquad e_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

Epipolar lines:

$\tilde{l}_2 = \mathcal{E}_1 \tilde{x}_1$

$$\tilde{l}_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ -1 \\ y \end{bmatrix}$$

$0 + (-1)Y + y = 0$

$-Y + y$

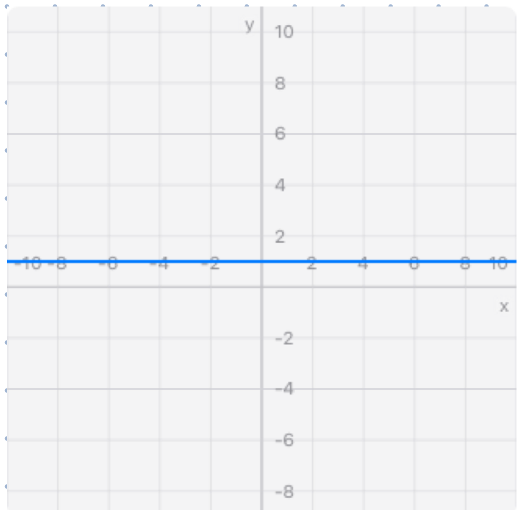
$Y = y$

So, y is fixed while x can vary.
This a horizontal line

$\tilde{l}_1 = \mathcal{E}_1^T \tilde{x}_2$

$$\tilde{l}_1 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 1 \\ -y \end{bmatrix}$$



$y = 1$

(iii) $\mathcal{E}_3 = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

NOTE: $[x]_\times = \begin{bmatrix} 0 & -z & y \\ z & 0 & -x \\ -y & x & 0 \end{bmatrix}$

Epipoles:

$\mathcal{E}_3 e_1 = 0$

$$\left[\begin{array}{ccc|c} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] R_2 \leftrightarrow R_1$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$\mathcal{E}_3^T e_1 = 0$

$$\left[\begin{array}{ccc|c} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] R_2 \leftrightarrow R_1$$

$$\left[\begin{array}{ccc|c} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$e_1 \begin{bmatrix}0\\0\\1\end{bmatrix}$

$e_2 = \begin{bmatrix}0\\0\\1\end{bmatrix}$

Epipolar lines:

$$\tilde{\ell}_2 = \mathcal{E}_3 x_1$$

$$= \begin{bmatrix}0 & -1 & 0\\1 & 0 & 0\\0 & 0 & 0\end{bmatrix}\begin{bmatrix}x\\y\\1\end{bmatrix}$$

$$\begin{bmatrix}-y\\x\\0\end{bmatrix}$$

$$-yX + xY = 0$$

$$Y = \frac{y}{x} X$$

$$\tilde{\ell}_1 = \mathcal{E}_3^T x_2$$

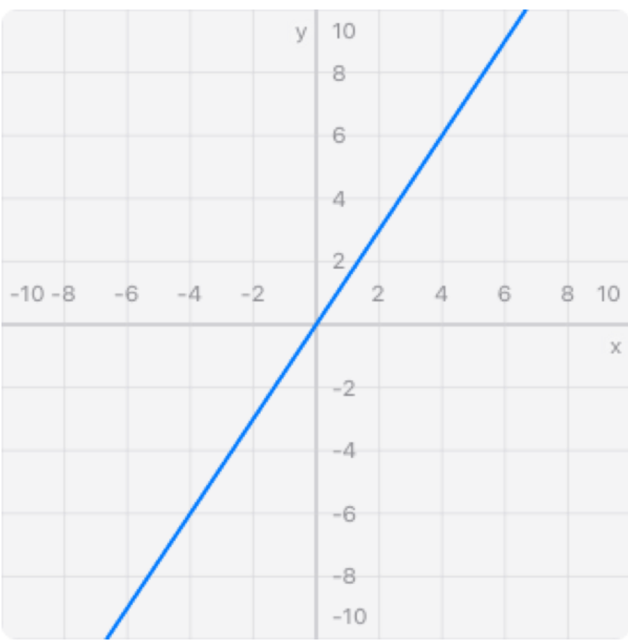
$$= \begin{bmatrix}0 & 1 & 0\\-1 & 0 & 0\\0 & 0 & 0\end{bmatrix}\begin{bmatrix}x\\y\\1\end{bmatrix}$$

$$\begin{bmatrix}y\\-x\\0\end{bmatrix}$$

$$yX - xY = 0$$

$$Y = \frac{y}{x} X$$

↳ slanted line



$y = 3,\ x = 2$

(c) $F = K_2^{-1} \mathcal{E} K_1^{-1}$

So, when $K_1 = K_2 = I$

$\tilde{x}_i = K_i \bar{x}_i$

$\tilde{x}_1 = \bar{x}_i$

Nothing happened

1.2

(a) $\bar{x}_i x \; P X$

(b) $P_1 c_1 = 0$

$K[R|t]c = 0$

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{3\times 3} \underbrace{\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}}_{3\times 4} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \underline{0}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = \underline{0}$$

$$\left.\begin{aligned} x &= 0 \\ y &= 0 \\ z + w &= 0 \end{aligned}\right\}$$

$z = -w$

$C_1 = \begin{bmatrix} 0 \\ 0 \\ -w \\ w \end{bmatrix}$ similarly, $t = \underline{0}$ $C_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ w \end{bmatrix}$

So,

$C_1 = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$ $C_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

So, baseline is lies principle axes.

Epipoles on principle point as $e = (0, 0)$

Q1.2:

(a)

$$P_1 = K_1[R_1|t_1]$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$P_2 = K_2[R_2|t_2]$$

$$= \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} -1 & 0 & 0 & -2 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & 0 & 1 & -3 \\ 0 & -2 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\tilde{X}_s \times P\tilde{X}_w = 0$$

$$[\tilde{X}_s]_\times P\tilde{X}_w = 0$$

NOTE: $[X]_\times = \begin{bmatrix} 0 & -z & y \\ z & 0 & -x \\ -y & x & 0 \end{bmatrix}$

$$\begin{bmatrix} 0 & -1 & 1/2 \\ 1 & 0 & -1/4 \\ -1/2 & 1/4 & 0 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \underline{0}$$

$$\begin{bmatrix} 0 & -1 & 1/5 \\ 1 & 0 & 1/5 \\ -1/5 & -1/5 & 0 \end{bmatrix}\begin{bmatrix} -2 & 0 & 1 & -3 \\ 0 & -2 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \underline{0}$$

$$\begin{bmatrix} 0 & -1 & 1/2 & 0 \\ 1 & 0 & -1/4 & 0 \\ -1/2 & 1/4 & 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \underline{0}$$

$$\begin{bmatrix} 0 & 2 & -4/5 & -4/5 \\ -2 & 0 & 6/5 & -14/5 \\ 2/5 & 2/5 & -2/5 & 2/5 \end{bmatrix}\begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = \underline{0}$$

Now,

$$\begin{bmatrix} 0 & -1 & 1/2 & 0 \\ 1 & 0 & -1/4 & 0 \\ -1/2 & 1/4 & 0 & 0 \\ 0 & 2 & -4/5 & -4/5 \\ -2 & 0 & 6/5 & -14/5 \\ 2/5 & 2/5 & -2/5 & 2/5 \end{bmatrix} \bar{x}_w = \underline{0}$$

$$\bar{x}_w = \begin{bmatrix} x_4 \\ 2x_4 \\ 4x_4 \\ x_4 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 2 \\ 4 \\ 1 \end{bmatrix}$$

WOW!

Q 1.3

(a) We know,

$$z = \frac{fb}{d}$$

$$\frac{\partial z}{\partial d} = \frac{\partial}{\partial d}\left(\frac{fb}{d}\right)$$

$$\Delta z = \frac{-fb}{d^2}\Delta d$$

$$= \frac{-fb}{\left(\frac{fb}{z}\right)^2}\Delta d \quad \because d = \frac{fb}{z}$$

$$= -fb \times \frac{z^2}{(fb)^2}\Delta d$$

$$= \frac{-z^2}{fb}\Delta d$$

So, Δz grows quadratically

(b)
- Increase the baseline
- Increase focal length

Disadvantages:

- As we increase the base,
  - the overlap decreases.
- As we increase the focal length,
  - Our FoV decreases
  - Blurring on the sides

Q1.4

(a)

$$w_i' = \alpha_i w + 1\beta_i$$

$$\frac{\left[\alpha_i w_L + 1\beta_L - \text{mean}\left(\{\alpha_i w_L + 1\beta_L\}\right)\right]\left[\alpha_R w_R + 1\beta_R - \text{mean}\left(\{\alpha_R w_R + 1\beta_R\}\right)\right]}{\text{Var}\left(\{\alpha_i w_L + 1\beta_L\}\right)\ \text{Var}\left(\{\alpha_R w_R + 1\beta_R\}\right)}$$

↳ We ignore β in Var

$$\frac{\left[\alpha_L w_L + \cancel{1\beta_L} - \alpha_L \text{mean}(\{w_L\}) - \cancel{1\beta_L}\right]\left[\alpha_R w_R + \cancel{1\beta_R} - \alpha_R \text{mean}(\{w_R\}) - \cancel{1\beta_R}\right]}{\text{Var}\left(\{\alpha_i w_L\}\right) \text{Var}\left(\{\alpha_R w_R\}\right)}$$

$$\frac{\cancel{\alpha_L}\left(w_L - \text{mean}(\{w_L\})\right)\ \cancel{\alpha_R}\left(w_R - \text{mean}(\{w_R\})\right)}{\cancel{\alpha_L}\,\text{std}(\{w_L\})\ \cancel{\alpha_R}\,\text{std}(\{w_R\})}$$

$$\frac{\left[w_L - \text{mean}(\{w_L\})\right]\left[w_R - \text{mean}(\{w_R\})\right]\alpha_R}{\text{std}(\{w_L\})\ \ \text{std}(\{w_R\})}$$

(b)

1	2	3	10	5	6	7
1	2	3	10	5	6	7
1	2	3	10	5	6	7
1	2	3	10	5	6	7
1	2	3	10	5	6	7

(a) Left image

2	10	4	5	6	7	8
2	10	4	5	6	7	8
2	10	4	5	6	7	8
2	10	4	5	6	7	8
2	10	4	5	6	7	8

(b) Right image

• $d=0$, $SAD = 3\times\left[\,|10-5| + |5-6| + |6-7|\,\right]$
$= 21$

1	2	3	10	5	6	7
1	2	3	10	5	6	7
1	2	3	10	5	6	7
1	2	3	10	5	6	7
1	2	3	10	5	6	7

(a) Left image

2	10	4	5	6	7	8
2	10	4	5	6	7	8
2	10	4	5	6	7	8
2	10	4	5	6	7	8
2	10	4	5	6	7	8

(b) Right image

• $d=1$, $3\times\left[\,|10-4| + |5-5| + |6-6|\,\right] = 18$

1	2	3	10	5	6	7
1	2	3	10	5	6	7
1	2	3	10	5	6	7
1	2	3	10	5	6	7
1	2	3	10	5	6	7

(a) Left image

2	10	4	5	6	7	8
2	10	4	5	6	7	8
2	10	4	5	6	7	8
2	10	4	5	6	7	8
2	10	4	5	6	7	8

(b) Right image

$d=2$, $3\times\left[\,|10-10| + |5-4| + |6-5|\,\right] = 3$

So, $d=2$ is best match but we can see [5] is at $d=1$.

(c)

1	0	1	2	2	2	1	0
2	0	1	2	2	2	1	0
3	0	1	2	2	2	1	0
4	0	1	2	2	2	1	0
5	0	1	2	2	2	1	0
	1	2	3	4	5	6	7

(a) Left → Right

1	2	2	2	2	1	0	0
2	2	2	2	2	1	0	0
3	2	2	2	2	1	0	0
4	2	2	2	2	1	0	0
5	2	2	2	2	1	0	0
	1	2	3	4	5	6	7

(b) Right → Left

Orange:

$d\_left(1,2)=1$ $\quad$ $d\_right(0,2)$ ✗

Cyan:

$d\_left(2,6)=1 \longrightarrow d\_right(2-1,6)=(1,6)$

$\downarrow$

$d\_right(1+0,6)=(1,6)$

Green:

$d\_left(3,4)=2 \longrightarrow d\_right(3-2,4)=(1,4)$ ✗

$d\_right(1,4)=2 \longrightarrow d\_left(1+2,4)=(3,4)$ back! ✓

Red

$d\_left(5,3)=2 \longrightarrow d\_right(5-2,3)=(3,3)$

$d\_right(3,3)=2 \longrightarrow d\_left(3+2,3)=(5,3)$ back ✓

Q1.5

(b)

- p_1: $c_\theta(d) = [1.0, 3.0, 10.0, 3.0, 1.0]$
- p_2: $c_\theta(d) = [10.0, 2.0, 1.0, 2.0, 10.0]$